



DIGITALMODEL VESSEL CAPABILITY GTM PACK · 2026-05-07

Deepwater Rigid Jumper Installation — Vessel Capability

Client-use note. Prepared as prospect-facing GTM collateral from representative engineering datasets and cited references. Class-typical vessel values are not a substitute for owner-approved vessel data, project-specific metocean, certified RAOs, or final installation engineering.

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Deepwater Rigid Jumper Installation Analysis

300 parametric cases across 2 vessels, 6 water depths, 5 jumper lengths, and 5 sea states

300 parametric cases analysed overnight

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Methodology

This analysis screens deepwater rigid jumper installation feasibility across a parametric matrix of vessels, water depths, jumper lengths, and sea states. Each case evaluates five installation phases:

Phase 1: Lift-off

Verifies the crane can lift the jumper plus rigging (5 te) with a **DAF of 1.10**. Hook load = (mass_per_m × length + rigging) × g × DAF. Checked against crane SWL at maximum radius.

Phase 2: In-air Bending (Jumper-Specific)

Rigid jumpers are long and slender — bending stress during lift is often the governing phase. The jumper is modelled as a **simply-supported beam** with uniform self-weight: $M = wL^2/8$, $\sigma = M \times (OD/2)/I$. Stress must not exceed **0.6 × SMYS** (268.8 MPa for X65).

Phase 3: Splash Zone

Hydrodynamic loads during passage through the air-water interface. Slamming force uses $C_s = \pi$ for circular pipe sections. Combined with drag and a **DAF of 1.30** for splash zone dynamics.

Phase 4: Lowering Through Water Column

Cable tension = submerged weight + cable weight + dynamic snap load. Dynamic load accounts for heave RAO and added mass ($C_a = 1.0$). Checked against **85% of wire MBL**.

Phase 5: Tie-in Alignment (Jumper-Specific)

After landing, the jumper must align to hub connectors within ± 50 mm. Current-induced lateral deflection is estimated using fixed-fixed beam theory with a deepwater current of 0.5 m/s.

Go/No-Go Classification

- GO:** all phases pass AND max utilisation < 0.85
- MARGINAL:** all phases pass AND max utilisation in [0.85, 1.00]
- NO_GO:** any phase fails (utilisation > 1.00)

Applicable Codes & Standards

- DNV-RP-H103 (2011) — Modelling and Analysis of Marine Operations
- DNV-ST-N001 (2021) — Marine Operations and Marine Warranty
- DNV-OS-F101 (2021) — Submarine Pipeline Systems (pipe stress limits)
- API 5L (2018) — Line Pipe (X65 material properties)

Summary: Installation Screening at 1500m Depth, Hs=2.0m

Parametric results for each vessel and jumper combination at reference conditions

VESSEL	JUMPER	LENGTH (M)	DEPTH (M)	HS (M)	STATUS	MAX UTIL	GOVERNING PHASE
Large CSV	Jumper-20m	20.0	1500	2.0	GO	0.315	In-air Bending
Large CSV	Jumper-40m	40.0	1500	2.0	NO_GO	2.964	Tie-in
Large CSV	Jumper-60m	60.0	1500	2.0	NO_GO	15.004	Tie-in
Large CSV	Jumper-80m	80.0	1500	2.0	NO_GO	47.420	Tie-in
Large CSV	Jumper-100m	100.0	1500	2.0	NO_GO	115.771	Tie-in

VESSEL	JUMPER	LENGTH (M)	DEPTH (M)	HS (M)	STATUS	MAX UTIL	GOVERNING PHASE
Medium CSV	Jumper-20m	20.0	1500	2.0	GO	0.315	In-air Bending
Medium CSV	Jumper-40m	40.0	1500	2.0	NO_GO	2.964	Tie-in
Medium CSV	Jumper-60m	60.0	1500	2.0	NO_GO	15.004	Tie-in
Medium CSV	Jumper-80m	80.0	1500	2.0	NO_GO	47.420	Tie-in
Medium CSV	Jumper-100m	100.0	1500	2.0	NO_GO	115.771	Tie-in

Take This Analysis Live During Operations

LIVE

This report used **design sea states** to screen parametric cases overnight. During the actual operation, **digitalmodel** can feed your vessel's measured motion data — VMMS, IMMS, MRU — directly into the same engineering models for **real-time go/no-go decisions**.

Instead of relying on forecasted Hs limits, jumper installation screening updates continuously with actual crane tip motions, hook loads, and dynamic amplification factors measured on your vessel.

\$500K VC-funded — we are building overnight engineering and real-time operations support exclusively for marine installation contractors. Early adopters get priority onboarding and custom model calibration for their fleet.

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Assumptions & Limitations

- All jumpers are 8" OD (219.1 mm) API 5L X65 with 18.26 mm wall thickness and FBE coating
- SMYS = 448 MPa, Young's modulus $E = 207 \text{ GPa}$
- Rigging mass = 5.0 te (slings, shackles, spreader bar)
- In-air bending modelled as simply-supported beam — conservative for typical dual-crane lifts
- Slamming coefficient $C_s = \pi$ for circular cross-section (DNV-RP-H103 Table 8-2)
- Drag coefficient $C_d = 1.2$ for cylinder (DNV-RP-H103)
- Added mass coefficient $C_a = 1.0$ for cylinder
- Lowering velocity = 0.5 m/s, typical for heavy-lift CSV operations
- Wire rope submerged weight = 50 N/m — representative of 96-128 mm diameter wire
- Deepwater current velocity = 0.5 m/s for tie-in deflection check
- Tie-in lateral tolerance = +/-50 mm (typical hub connector specification)
- T_p estimated from H_s via $T_p = 4.0 \times \sqrt{H_s}$ — simplified JONSWAP relationship
- Vessel RAOs are simplified single-peak values, not full frequency-dependent transfer functions
- No wave-current interaction or VIV effects during lowering
- No seabed slope or landing zone geotechnical effects considered

At Hs = 2.0 m. Cell shows governing phase and utilisation. Green = GO, Amber = MARGINAL, Red = NO_GO.

Medium CSV

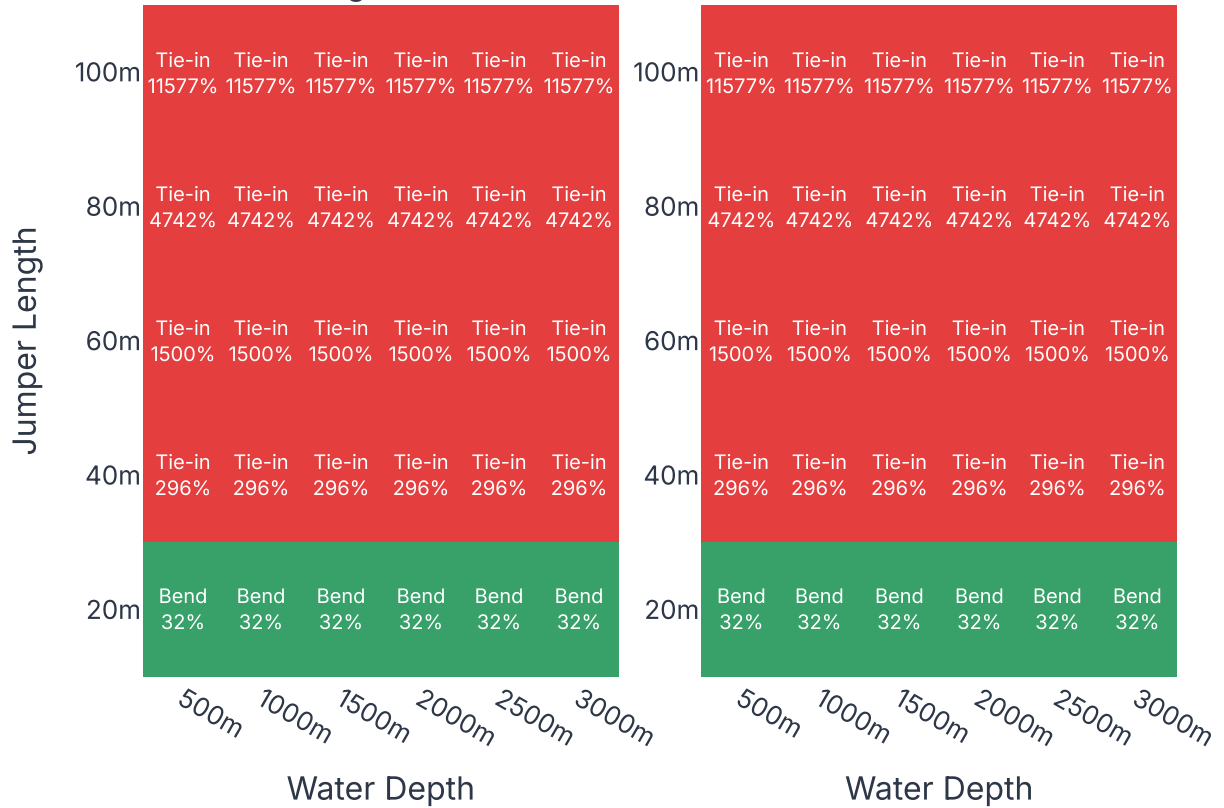


Chart 2: In-Air Bending Stress vs Jumper Length

Simply-supported beam model. Red dashed line = $0.6 \times \text{SMYS allowable}$ (268.8 MPa).

In-Air Bending Stress vs Jumper Length

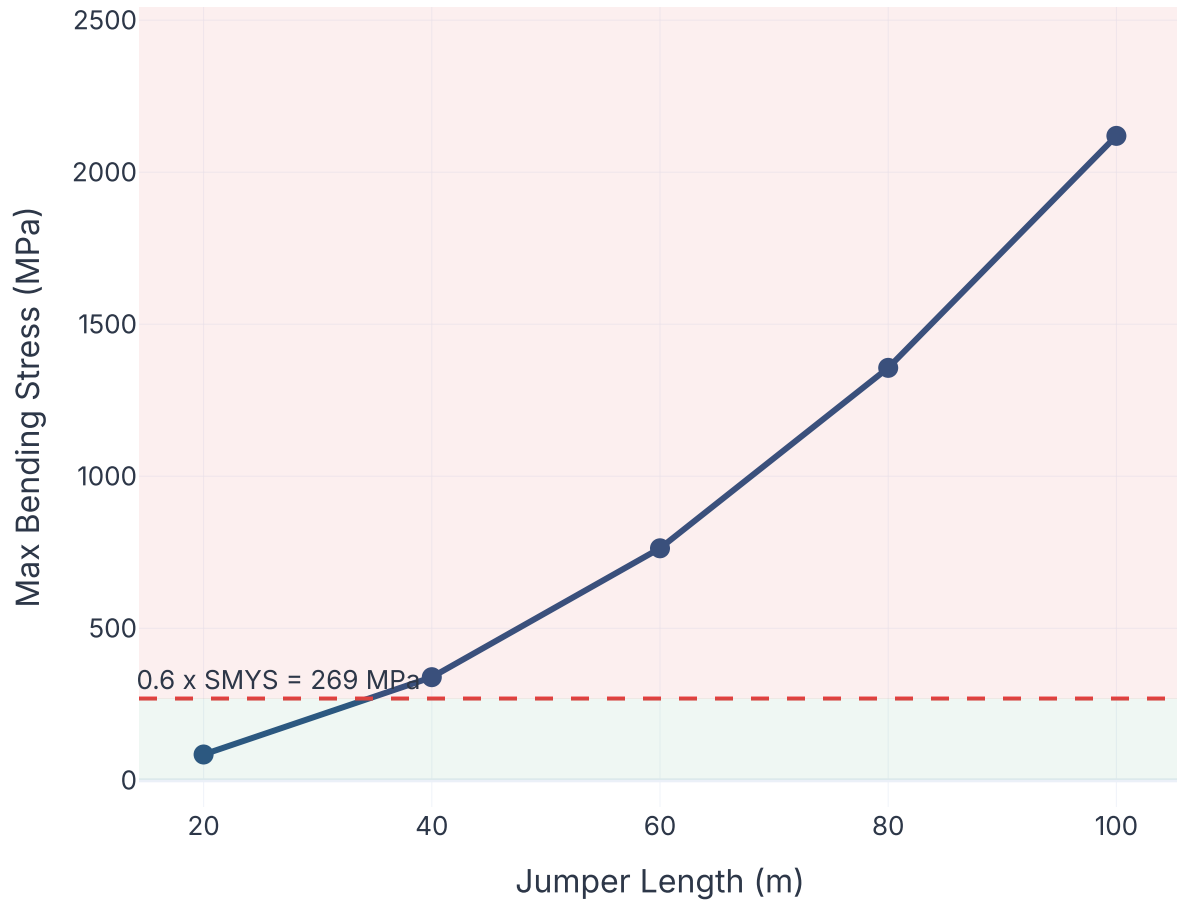


Chart 3: Operability Envelope — Max Allowable Hs

Maximum sea state for each jumper length at 1500 m water depth. One curve per vessel.

Operability Envelope — Max Allowable Hs at 1500m Depth

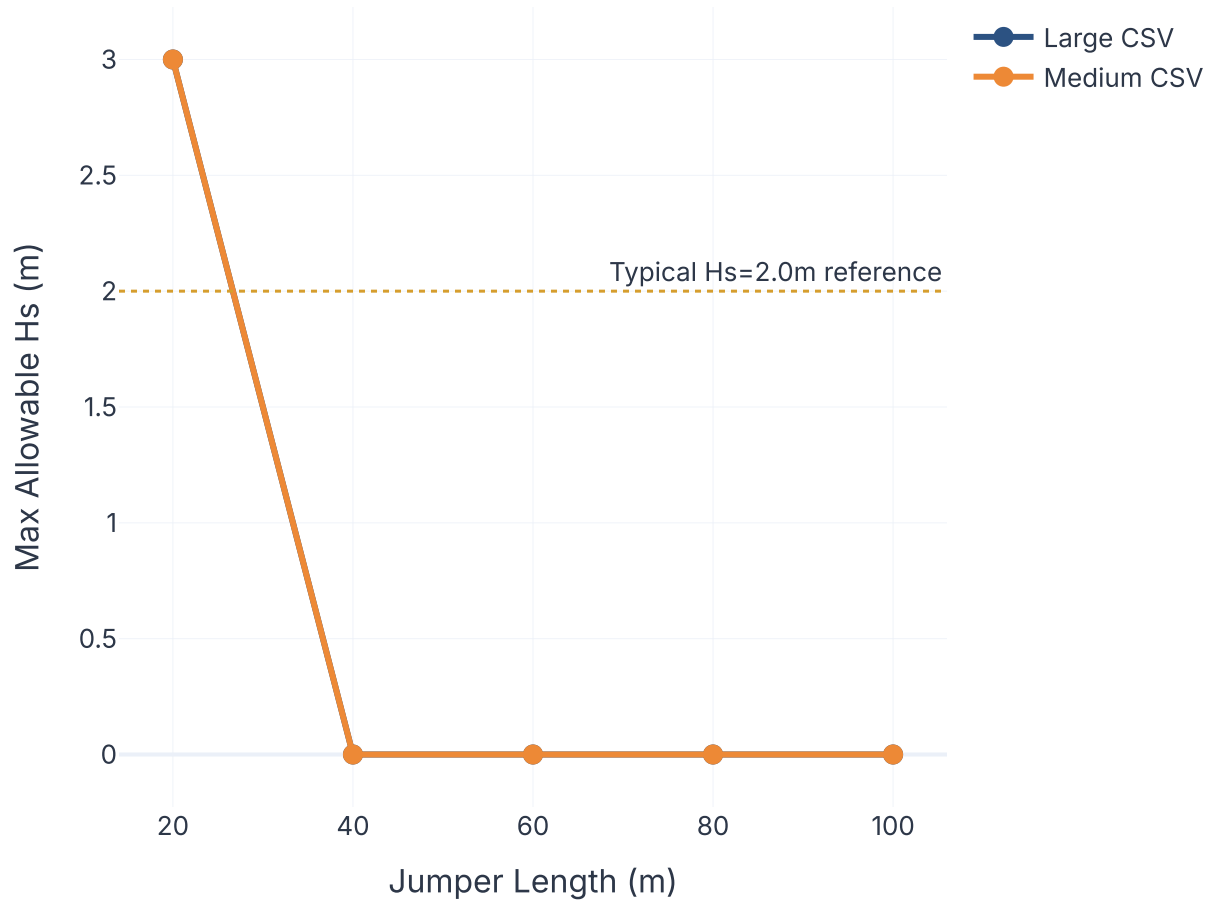


Chart 4: Cable Tension vs Water Depth

Large CSV at Hs = 2.0 m. Lines per jumper length. Red = 85% wire MBL.

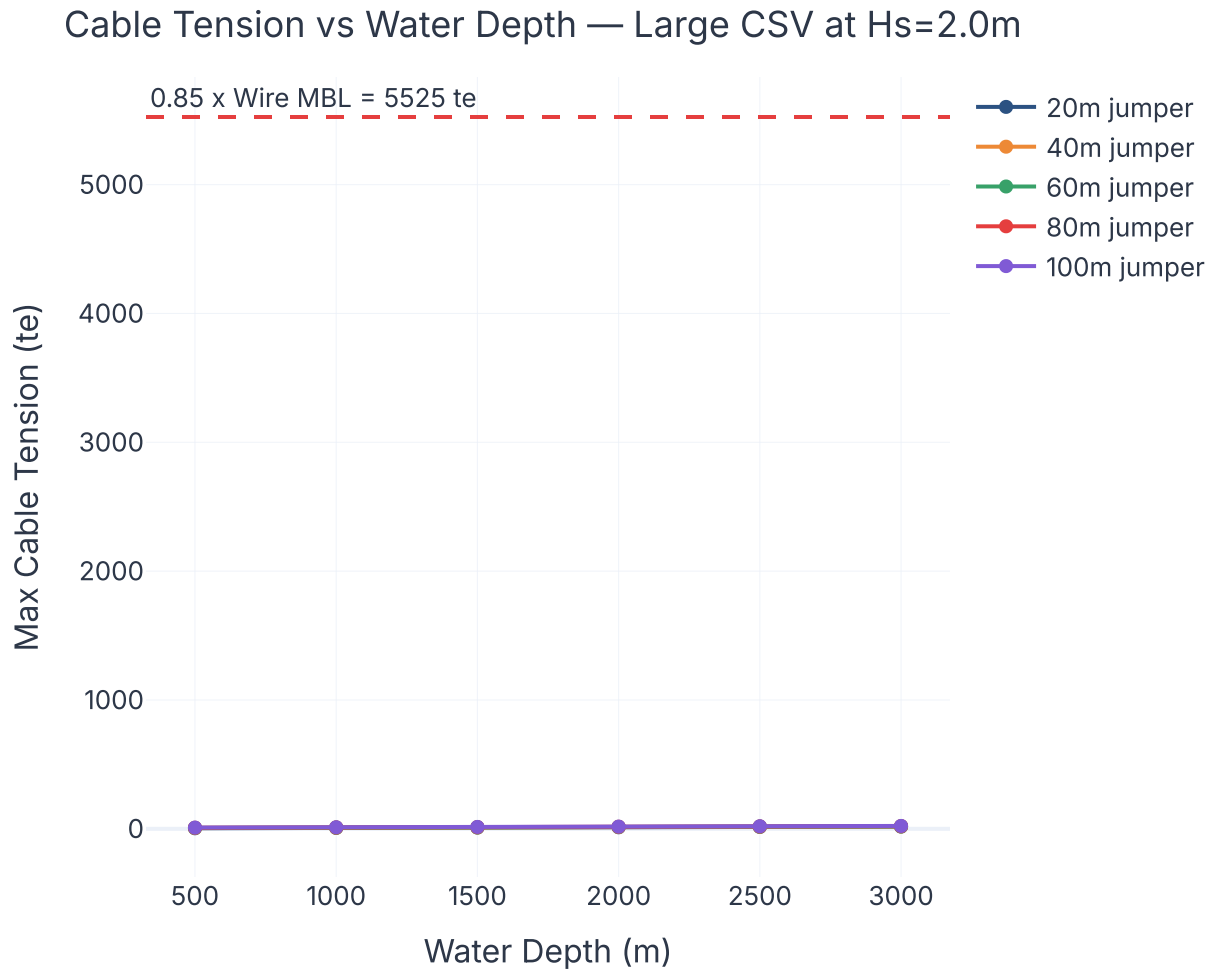
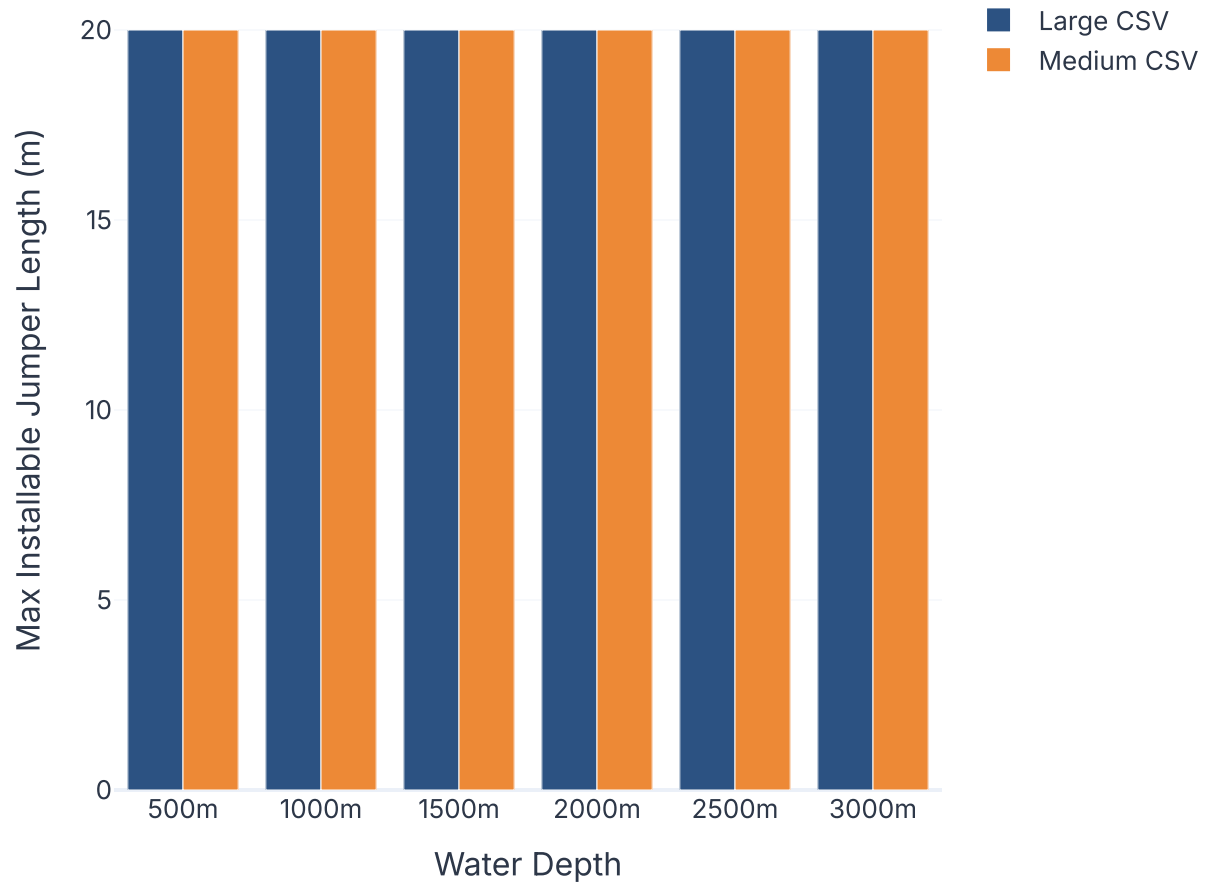


Chart 5: Vessel Comparison — Max Installable Jumper Length

At $H_s = 2.0$ m. Grouped by water depth.

Max Installable Jumper Length by Vessel at $H_s=2.0$ m



These results are preliminary engineering estimates generated by automated parametric analysis. All outputs require review by a qualified engineer before use in design or operational decisions.

References: DNV-RP-H103 (2011) — Modelling and Analysis of Marine Operations · DNV-ST-N001 (2021) — Marine Operations and Marine Warranty · DNV-OS-F101 (2021) — Submarine Pipeline Systems (pipe stress limits) · API 5L (2018) — Line Pipe (X65 material properties)

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